Parametric & Sequences Test, Day 1

Cram sheet

Wednesday, no calculator: 5 parametric curve questions + 1 sequence question (terms, recursive or Σ notation). 16 points.

How far to simplify (every rectangular answer)

- **Whole graph shows** (circle, ellipse, both branches): done once t is gone, e.g. $\frac{x^2}{4} - y^2 = 1$. Solving for y is also right, only with $\pm$.
- **Part of it, a domain can cut it** (piece of a parabola, one left-right branch): add "for x in $[a,b]$", e.g. $\frac{x^2}{4} - y^2 = 1$ for x in $[2,\infty)$.
- **Part of it, a domain can't** (top or bottom of an up-down hyperbola): solve for y, keep one sign, e.g. $y = 2\sqrt{\frac{x^2}{9} + 1}$.
- **Small piece of a branch:** solve for y *and* add "for x in". Never "for y in" or "$y \geq$".
- Question says **SIMPLIFIED**: all the way, y alone ($y = -\frac{1}{3}x^2$). No "simplify": any form with t gone; don't multiply out.

Parametric curves

1. **t-chart first.** Brackets: plug in both ends, solid dots. Parentheses: values inside.
2. Solve for t where it gives **one answer** (no $\pm$); substitute into the other.
3. Domain: smallest and largest x in the chart, "for x in $[a,b]$". Watch turning points ($x = t^2 \geq 0$, $x = \sqrt{t} \geq 0$).
4. Any form is fine once t is gone; no multiplying out. **SIMPLIFIED** means y alone. No domain written means all real x.
5. Direction arrows **only when the question asks**: lowest t to highest. End arrowheads only when the curve keeps going. Asymptotes optional, dashed if drawn.

What the t-chart shows	**Final form** (whole graph: done; domain can cut it: "for x in"; can't: solve for y, one sign)
full circle / ellipse	$\frac{x^2}{9} + \frac{y^2}{25} = 1$, no domain
one hyperbola branch (restricted t)	solve for y, keep that branch's sign ($+\sqrt{}$ top, $-\sqrt{}$ bottom): $y = 2\sqrt{\frac{x^2}{9} + 1}$; never "$y \geq$"; any equivalent form
both branches (no restriction)	$\frac{x^2}{4} - \frac{y^2}{9} = 1$, done (or $y = \pm 3\sqrt{\frac{x^2}{4} - 1}$, never $+\sqrt{}$ alone)
one left-right branch	$\frac{x^2}{4} - y^2 = 1$ for x in $[2,\infty)$; never "for y in"
reciprocal $x = t+2,\ y = \frac{1}{t+2}$	$y = \frac{1}{x}$; "for all $x \neq 0$" is good practice (it comes with the equation)

Two parametric representations

Give both $x =$ and $y =$ ($t = 4 - x$ is not parametric form). $y = (x-2)^2 + 1$: $x = t,\ y = (t-2)^2 + 1$ and $x = t+2,\ y = t^2 + 1$, for t in $(-\infty,\infty)$ (or the restricted interval). Not in vertex form? Pick any shift $x = t + h$, substitute, simplify.

Identities for eliminating t

$\sin^2 t + \cos^2 t = 1$, $1 + \tan^2 t = \sec^2 t$, $1 + \cot^2 t = \csc^2 t$.

The sequence question (one of these)

- **First five terms:** plug in $n = 1,2,3,4,5$. Answer is a list with commas, reduced, no ellipsis. $(-1)^n$: odd n negative.
- **Recursive:** the formula starts where it says ($n \geq 2$, or $n \geq 3$ when two terms are given); a_{n-1} is the previous term, a_{n-2} the one before it.
- **Write out a sum:** start at the lower index (it can be $k = 0$), first few terms, ellipsis, last term with n; don't add.
- **Express as Σ:** lower index, upper index, general term, same letter. Check the first and last terms.

Ex: $a_1 = -1,\ a_2 = 3,\ a_n = a_{n-1}a_{n-2} \to -1, 3, -3, -9, 27$ $\quad \frac{1}{3} + \frac{2}{4} + \cdots + \frac{15}{17} = \sum_{k=1}^{15} \frac{k}{k+2}$

Where points get lost

- Writing only $+\sqrt{}$ for a whole hyperbola (leave it in "$= 1$" form or keep $\pm$).
- Arithmetic slips while simplifying, like $\left(\frac{x+3}{-2}\right)^2$ with -2 on the bottom instead of 4: check one t-chart point in the final equation.
- Recursive: starting the formula at the wrong n; Σ: starting at 1 when it says $k = 0$.